SANTOSH (RUSTAM) SHRESTHA

Machine Learning Engineer & Software Engineer (Full-Stack & Applied ML)

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TECHNICAL SKILLS

Languages:	JavaScript (ES6+), Python, TypeScript, PHP, SQL, HTML5, CSS3, Java, C, C#
Frontend Engineering:	React, Redux Toolkit, TanStack Query, shadcn/ui, Tailwind CSS, Zustand, Vite, Responsive Web Design
Backend & APIs:	PHP, Laravel, Eloquent ORM, Composer, FastAPI, Node.js, Express.js, RESTful APIs, WebSockets / Socket.io
Databases & Storage:	PostgreSQL, MySQL, MongoDB, Redis, ChromaDB (Vector Store for RAG), Index Tuning & Idempotency
Machine Learning:	Classical ML (PCA, LDA, Random Forest, Logistic/Linear Regression, SVM), XGBoost, scikit-learn, PyTorch, Transformers, RAG & ChromaDB, Agentic GenAI, Pandas, NumPy, SHAP, OpenCV
DevOps & Tooling:	GitHub Actions, CI/CD YAML, Docker, DVC, MLOps, Postman, Git, GitHub, WAMP, Flyenv, Python Locust

PROFESSIONAL EXPERIENCE

JYRA Soft (YBC ERP) | **Full-Stack MERN Developer (ERP Systems)** **Dec 2025 - Present | Kathmandu, Nepal**

- **Re-engineered** legacy codebase from prop-drilling hurdles into modular React components with Redux Toolkit slices and TanStack Query caching, reducing redundant database roundtrips by 35-40%.
- **Built** end-to-end timesheet tracking, periodic task scheduling, dynamic assignments, leave management, and resignation day-off workflows with normalized client state.
- **Engineered** accounting & financial modules (staff remuneration, allowances, accountant dashboards, budget setting, payroll) with fine-grained RBAC/ABAC permission controls.
- **Implemented** reusable Express middlewares resolving duplicate requests and floating-point bottlenecks; integrated Redis caching and hash sets for session sync and endpoint latency reduction.

Saurabh Group (Sarbottam & Laxmi Steels) | **Full-Stack Developer** **Apr 2025 - Jul 2025 | Kathmandu, Nepal**

- **Migrated** legacy manual raw SQL joins and nested loops to Laravel Eloquent ORM with eager/lazy loading, increasing database query throughput by 35% across 300K-500K+ active records.
- **Built** industrial logistics, scrap in/out quality tracking, and weighment systems with audit logs for workers, admins, and directors handling hundreds of daily steel transactions.
- **Developed** authorized agent systems with PAN validation and long-term booklet sales tracking with date-indexed query optimizations; containerized with Docker and Flyenv on WAMP.
- **Profiled** database throughput and latency using Python Locust; offloaded heavy validation to the frontend to minimize server load.

Kath Khabar Media (काठ खबर) | **Graphic Designer & Concept Artist** **Nov 2024 - Mar 2025 | Lalitpur, Nepal**

- **Produced** routine publication graphics, breaking news banners, and YouTube thumbnails under rapid publication deadlines using Photoshop and Microsoft Designer.
- **Created** visual concept campaigns for cultural events including Shivaratri, Saraswati Puja, and International Women's Day.

SELECTED PROJECTS

FinGuard: Multi-Tenant Loan & ML Risk System | PERN Stack, FastAPI, XGBoost, Docker, SHAP **github.com/Rustam-Shrestha/el-proyecto-de-anos-final**

- **Architected** multi-tenant PERN loan platform (4 actor roles); served real-time credit-risk inferences via FastAPI microservice running an XGBoost model evaluated with cost-weighted threshold (0.3748) trained on 250K-300K+ records, delivering 85.01% holdout accuracy (0.7630 AUC-ROC) and SHAP explainability.

SnapAttendance: Biometric Face & Geofenced Attendance | MTCNN, HOG, RBF SVM, FastAPI, PHP, MySQL, Haversine GPS **github.com/Rustam-Shrestha/snapattendance_reengineered**

- **Built** biometric pipeline with MTCNN 5-landmark face detector, head-pose gating, 15,876-dim HOG extraction, and RBF-kernel SVM (5-fold CV grid search); implemented 3-pose confirmation (>0.75 confidence) and Haversine GPS geofencing endpoint validating attendance within 200m (0.2 km).

DataSphere: Structured Data Intelligence & Schema Engine | Python, FastAPI, React, PostgreSQL, Gemini API **github.com/Rustam-Shrestha/DataSphere**

- **Built** full-stack tabular data ingestion engine for CSV, XLS, and XLSX spreadsheets into PostgreSQL with Gemini API natural language query translation and interactive analytics dashboards handling 5,000-10,000+ data points.

AI Lyric & Next-Word Sequence Generator | PyTorch, LSTM, NLP, FastAPI, NLTK **github.com/Rustam-Shrestha/deep_learning_for_text_pytorch**

- **Engineered** PyTorch LSTM language model trained on lyric_dataset.txt corpus with NLTK tokenization, learned word embeddings, temperature-controlled sampling, and low-latency FastAPI inference server.

EDUCATION

Bachelor in Computer Application (BCA) | Saraswati Multiple Campus, Tribhuvan University **2022 - 2026 | Kathmandu, Nepal**

- **Relevant Coursework:** Database Management Systems (DBMS), Data Structures & Algorithms (DSA), Java, Web Technologies, Operating Systems, Software Engineering, AI & Machine Learning.

CERTIFICATIONS

Full-Stack & Systems: Full-Stack Web Development (YouAccel), Web Designing (Global IT), MLOps Fundamentals (DataCamp).
Applied Machine Learning: Math for ML (Udemy), Deep Learning with PyTorch (DataCamp), XGBoost (DataCamp), Explainable AI with SHAP (DataCamp).